

HR ANALYTICS PROJECT

Employee Attrition & Workforce Insights

A practical data analytics project using Excel, PostgreSQL and Power BI

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1. Project Overview

This project is about analysing employee information and understanding the factors that are associated with employee attrition. I created and cleaned a 2,000-record HR dataset and used three tools at different stages of the project: Microsoft Excel for initial cleaning and exploration, PostgreSQL for structured data storage and SQL analysis, and Power BI for the final interactive dashboard.

My main aim was not only to calculate attrition, but also to look at attrition from different HR angles such as department, job role, overtime, salary band, job satisfaction and location. The final dashboard brings these results together so that an HR person can quickly identify areas that may need attention.

2. Abstract

Employee attrition is an important HR problem because frequent employee exits can affect productivity, workload, recruitment cost and team stability. In this project, I analysed a synthetic HR employee dataset containing 2,000 employee records. I first cleaned and checked the data in Excel, then imported the cleaned data into PostgreSQL and performed SQL analysis. I also created SQL views and used CTEs and window functions for deeper analysis. Finally, I connected PostgreSQL to Power BI and built an interactive HR Analytics Dashboard.

The overall observed attrition rate in the dataset is 14.90%, with 298 employees marked as having left. Finance has the highest department attrition at 17.31%. Employees working overtime have a higher observed attrition rate of 19.77% compared with 12.29% for employees without overtime. Customer Service Executive has the highest observed job-role attrition at 25.00%. The ₹40K-₹60K salary band has the highest salary-band attrition at 16.07%, while job satisfaction level 2 has the highest observed attrition at 20.66%.

These results show areas for HR investigation and retention planning. They are observed patterns in the dataset and should not be interpreted as proof that one factor directly causes attrition.

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4. Introduction

Human resources is not only about recruitment and employee administration. Today, HR teams also use data to understand workforce behaviour and make better decisions. One of the useful areas for HR analytics is employee attrition. If a company can identify departments, roles or working conditions where attrition is comparatively high, HR can investigate the reasons and plan suitable retention actions.

I selected this topic because it combines my interest in HR with data analysis. The project gave me an opportunity to work through a complete analytics process instead of only creating a dashboard. I worked with the data from the initial cleaning stage, then moved to database analysis and finally presented the results in Power BI.

5. Problem Statement

The main problem addressed in this project is to understand employee attrition patterns from an HR dataset. The analysis focuses on questions such as:

- What is the overall employee attrition rate?
- Which departments have higher attrition?
- Which job roles have higher attrition?
- Is attrition different between employees with and without overtime?
- Which salary bands show higher attrition?
- How does job satisfaction relate to observed attrition?
- Which roles appear to be high-risk within each department?
- How can HR use these findings for retention planning?

6. Objectives

- Clean and prepare the HR employee dataset for analysis.
- Check data quality issues such as blanks, duplicates, incorrect data types and unusual salary values.
- Store the cleaned data in PostgreSQL.
- Use SQL to calculate workforce and attrition metrics.
- Use CTEs and window functions to perform deeper role and department analysis.
- Create reusable SQL views for Power BI.
- Build an interactive Power BI dashboard with KPIs, charts, slicers and a location map.
- Identify practical HR insights and recommendations from the observed patterns.
- Present the complete project as a portfolio piece for data analytics and HR-related job opportunities.

7. Dataset Description

The project uses a synthetic HR employee dataset with 2,000 employee records. The data was created for learning and portfolio purposes, so it should not be treated as confidential company data or as a representation of a real organisation.

Column	Description
employee_id	Unique employee identifier
employee_name	Employee name
gender	Employee gender category
age	Employee age
department	Department of the employee
job_role	Employee job role
location	Employee work location
joining_date	Date the employee joined

experience_years	Years of experience
salary_inr	Salary in Indian rupees
performance_score	Employee performance score
attendance_percentage	Attendance percentage
overtime	Whether the employee works overtime
training_hours	Training hours
job_satisfaction	Job satisfaction rating
work_hours_per_week	Weekly working hours
promotion_last_3_years	Promotion status in last three years
attrition	Whether the employee left
exit_date	Exit date, where applicable

8. Tools and Technologies Used

Tool	Purpose
Microsoft Excel	Data cleaning, descriptive statistics, outlier checking and pivot analysis
PostgreSQL	Database creation, data storage and SQL analysis
SQL	Aggregations, CTEs, window functions and reusable views
Power BI	Interactive dashboard, KPIs, charts, slicers and map

9. Project Methodology

1. Dataset creation and initial inspection
2. Excel cleaning and quality checks
3. Descriptive and pivot analysis
4. PostgreSQL database and table creation
5. CSV import and data-type corrections
6. Basic SQL analysis
7. Advanced SQL using CTEs and window functions
8. Creation of SQL views
9. Power BI connection and data-type validation
10. DAX measure creation
11. Dashboard design and interaction testing
12. Key insights, recommendations and final documentation

10. Data Cleaning and Preparation

Before doing the main analysis, I cleaned the dataset in Excel. This stage was important because incorrect values can affect the final SQL results and Power BI visuals.

10.1 Cleaning activities completed

- Checked the dataset structure and column names.
- Handled blank values in fields such as Attendance, Training, Job Satisfaction and Location.
- Standardised location naming by using 'Bengaluru' instead of 'Bangalore'.
- Checked salary values for unusual observations.
- Calculated salary statistics and used the IQR method for outlier checking.
- Identified one low salary outlier and 43 high salary outliers.
- Reviewed the low outlier EMP1612 and replaced the salary value of ₹18,375 with ₹60,500 after checking the role/context.
- Saved the cleaned dataset as 'HR_Analytics_cleaned_data finished.csv'.

10.2 Salary statistics

Measure	Value
Minimum salary	₹18,375
Maximum salary	₹203,000
Q1	₹52,500
Median / Q2	₹69,500
IQR	₹17,000
Upper-bound reference used	₹44,000 (as recorded during project analysis)
Low outliers	1
High outliers	43

Note: the salary outlier-check values above are recorded from the project working notes. The IQR rule normally uses $Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$, so the bound should be rechecked before using it as a formal statistical statement in an academic submission.

11. Excel Analysis

Excel was my first analysis stage. I used it to understand the dataset before moving to PostgreSQL. This helped me identify data-quality issues early and gave me a basic understanding of the workforce structure.

11.1 Department employee distribution

Department	Employees
Finance	260
Customer Service	189
Marketing	221
HR	278
Sales	406
Operations	297
IT	349

Sales is the largest department in the dataset with 406 employees, while Customer Service has the smallest department size with 189 employees.

11.2 Average salary by job role

Job role	Count	Average salary
Accountant	62	₹63,451.61
Business Development Executive	99	₹58,808.08
Customer Service Executive	40	₹47,350.00
Customer Service Manager	54	₹50,611.11
Customer Support Analyst	44	₹48,988.64
Data Analyst	72	₹90,479.17
Digital Marketing Executive	55	₹61,354.55
Finance Executive	76	₹65,677.63
Finance Manager	58	₹63,413.79
Financial Analyst	64	₹71,367.19
HR Analyst	66	₹55,515.15
HR Executive	57	₹54,657.89
HR Manager	77	₹57,941.56
IT Manager	80	₹76,168.75
IT Support Executive	92	₹70,755.43
Marketing Analyst	62	₹59,370.97
Marketing Executive	54	₹59,647.69
Marketing Manager	50	₹59,870.00
Operations Analyst	74	₹58,581.08
Operations Executive	69	₹60,340.58
Operations Manager	74	₹59,153.72
Process Associate	80	₹53,150.00
Recruiter	78	₹54,134.62
Sales Analyst	107	₹56,953.27

Sales Executive	96	₹57,854.17
Sales Manager	104	₹56,783.65
Software Analyst	105	₹84,492.14
Team Leader	51	₹61,049.02
Grand Total	2,000	₹62,146.51

12. PostgreSQL Database and SQL Analysis

After cleaning the CSV, I created an employees table in PostgreSQL and imported 2,000 records. During the import stage, I faced some practical errors, which helped me understand how CSV structure and PostgreSQL data types need to match.

12.1 Import issues handled

- Extra data after the last expected column because the CSV contained an additional Salary_Outlier_Check field.
- Blank exit_date values causing date conversion errors.
- Incorrect NULL specification during CSV import.
- Decimal training_hours values such as 29.74814815 could not fit into an integer column.
- The training_hours column was therefore changed to DECIMAL(10,2).

After the fixes, the complete 2,000-row dataset was successfully imported. I also checked employee_id duplicates and received 0 duplicate rows.

12.2 Basic SQL analysis

The main SQL analysis calculated department size, average salary, attrition counts and attrition rates. I also grouped results by overtime, job satisfaction, salary band and job role.

Overall attrition was calculated as 14.90%, based on 298 employees who had attrition = 'Yes' out of 2,000 employees.

13. Advanced SQL Analysis

To make the project stronger from a SQL point of view, I used Common Table Expressions (CTEs) and window functions. These helped me compare departments and roles against the company benchmark instead of looking only at simple totals.

13.1 CTE analysis completed

- Department attrition calculation
- Department results compared side-by-side with the company benchmark
- Departments above the company average attrition
- Risk classification
- Job roles above company average attrition
- Job roles above company average with at least 50 employees

13.2 Window functions completed

- RANK() to rank job roles by attrition rate
- LAG() to compare the current role with the previous role
- Difference between current and previous attrition rates
- Running average of role attrition
- ROW_NUMBER() to identify the top 5 roles

- PARTITION BY department to rank roles within each department

13.3 Highest-risk role in each department

Department	Highest-risk role	Employees	Left	Attrition	Vs company
Customer Service	Customer Service Executive	40	10	25.00%	+10.10 pp
Finance	Finance Manager	58	14	24.14%	+9.24 pp
Marketing	Marketing Executive	54	13	24.07%	+9.17 pp
HR	HR Analyst	66	14	21.21%	+6.31 pp
IT	Software Analyst	105	20	19.05%	+4.15 pp
Operations	Process Associate	80	14	17.50%	+2.60 pp
Sales	Sales Executive	96	16	16.67%	+1.77 pp

The company benchmark used for this comparison is 14.90%. The difference column shows the role's observed attrition rate above or below that benchmark.

14. SQL Views

I created five PostgreSQL views so that the final Power BI dashboard could use clean, reusable analysis outputs where appropriate.

- vw_department_attrition - department-level attrition summary
- vw_job_role_attrition - job-role attrition summary
- vw_salary_attrition - salary-band attrition summary
- vw_overtime_attrition - overtime attrition summary
- vw_high_risk_roles - high-risk role analysis

15. Power BI Dashboard

After completing the SQL work, I connected PostgreSQL to Power BI. I checked the data types before creating the visuals. The final dashboard was designed to give a quick HR-level view of workforce size, attrition and possible risk areas.

15.1 DAX measures created

```
Total Employees = COUNTROWS(employees)
Employees Left = CALCULATE(COUNTROWS(employees), employees[attrition] = "Yes")
Attrition Rate = DIVIDE([Employees Left], [Total Employees], 0)
Average Salary = AVERAGE(employees[salary_inr])
Average Attendance = AVERAGE(employees[attendance_percentage])
```

15.2 Dashboard KPIs

KPI	Result
Total Employees	2,000
Employees Left	298
Attrition Rate	14.90%
Average Salary	Approximately ₹62.15K
Average Attendance	91.88

15.3 Dashboard visuals

- Department Attrition - donut chart showing the attrition rate for each department.
- Overtime Attrition - donut chart comparing employees with and without overtime.
- Employee Distribution by Location - map showing employee locations.

- Salary Band Attrition - column chart comparing attrition across salary bands.
- Job Role Attrition - horizontal bar chart highlighting high-attrition roles.
- Job Satisfaction Attrition - line chart showing the attrition rate for satisfaction levels 1 to 5.
- Six compact Key HR Insight cards at the bottom of the dashboard.

15.4 Final Dashboard Screenshot

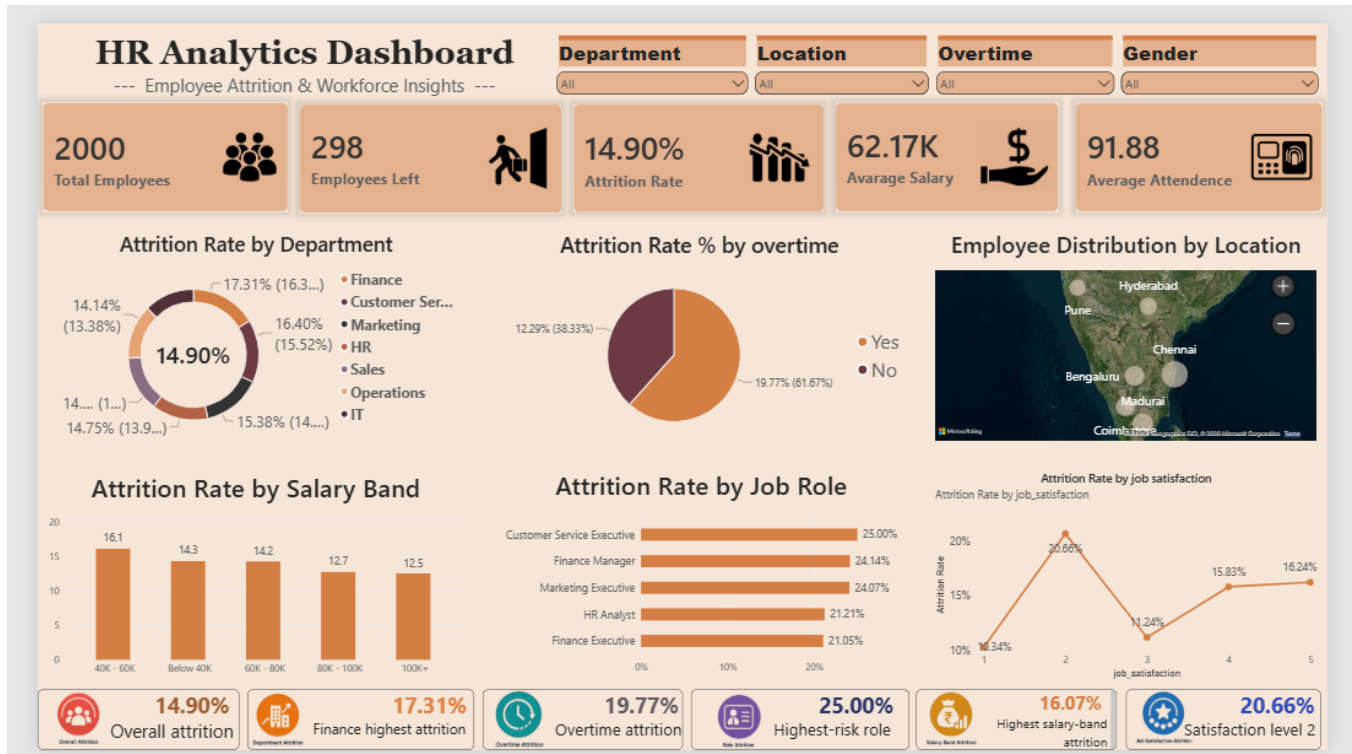


Figure 1: Final HR Analytics Dashboard

16. Key Findings

Overall attrition: 14.90% of employees left the organisation in the dataset, equal to 298 employees.

Department: Finance has the highest department attrition at 17.31%, followed by Customer Service at 16.40%.

Overtime: Employees with overtime have an observed attrition rate of 19.77%, compared with 12.29% for employees without overtime.

Job role: Customer Service Executive has the highest observed job-role attrition at 25.00%.

Salary band: The ₹40K-₹60K band has the highest observed salary-band attrition at 16.07%.

Job satisfaction: Satisfaction level 2 has the highest observed attrition at 20.66%.

Department-role risk: Each department has a highest-risk role above the company benchmark, with Customer Service Executive highest overall at 25.00%.

Attendance: Average attendance in the dashboard is 91.88.

17. HR Recommendations

Reduce overtime pressure - Review workload, staffing levels and overtime allocation because the overtime group shows a higher observed attrition rate.

Focus on Finance - Finance has the highest department attrition, so HR can conduct retention discussions and identify department-specific issues.

Target high-risk roles - Customer Service Executive shows the highest observed role attrition. HR can review workload, career growth, compensation and employee engagement for this role.

Review the ₹40K-₹60K salary band - This band has the highest observed salary-band attrition, so compensation competitiveness and career progression can be reviewed.

Improve employee satisfaction - Satisfaction level 2 has the highest observed attrition. Employee feedback and engagement activities can be prioritised for employees showing lower satisfaction.

Monitor attrition regularly - HR can build a recurring dashboard review around department, role, overtime, salary and satisfaction so that potential retention risks are noticed early.

Important interpretation note: the recommendations are based on observed patterns in the dataset. They identify areas for HR investigation, but they do not prove that a particular factor directly causes employees to leave.

18. Project Limitations

- The dataset is synthetic and created for portfolio/learning purposes.
- The analysis shows association and patterns, not causal relationships.
- The project does not include qualitative information such as exit interview comments.
- Some categories have smaller employee counts, so percentages should be interpreted together with the number of employees.
- The dashboard is based on the available fields in the dataset and does not include external market salary or industry benchmark data.

19. Future Scope

- Add exit interview or employee feedback data for qualitative analysis.
- Build predictive attrition modelling using Python or another machine-learning tool.
- Add tenure bands and promotion analysis to study career progression.
- Compare salary against external market benchmarks.
- Create automated data refresh and scheduled HR reporting.
- Add a dedicated manager-level or department drill-through page in Power BI.

20. Conclusion

This project helped me understand how a complete data analytics workflow can be applied to an HR problem. I started with raw employee data, cleaned and checked it in Excel, stored and analysed it in PostgreSQL, and then converted the analysis into an interactive Power BI dashboard.

The main result from the project is that the dataset has a 14.90% observed attrition rate. Finance has the highest department attrition, overtime employees show higher attrition than non-overtime employees, Customer Service Executive has the highest observed role attrition, the ₹40K-₹60K band has the highest salary-band attrition, and satisfaction level 2 has the highest observed attrition among the satisfaction groups.

Personally, this project gave me practice in Excel, SQL, database handling, DAX and Power BI. More importantly, it helped me understand how to convert numbers into HR-related business questions and recommendations.

Thank you